

HARKAMAL SINGH TOOR

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DATA SCIENTIST | MACHINE LEARNING | EXPERIMENTATION

Data Scientist with hands-on experience building end-to-end machine learning, experimentation, causal inference, forecasting, and decision-support projects using Python and SQL. Strong foundation in statistical analysis, A/B testing, feature engineering, model evaluation, and translating analytical results into business decisions. Built production-oriented ML systems incorporating model tracking, APIs, Docker, monitoring, automated testing, and interactive analytics applications.

TECHNICAL SKILLS

Languages: Python, SQL

Data Science & Machine Learning: pandas, NumPy, scikit-learn, CatBoost, XGBoost, LightGBM, Feature Engineering, Model Evaluation, Classification, Regression, Time Series Forecasting

Statistics & Experimentation: Statistical Inference, Hypothesis Testing, A/B Testing, Power Analysis, Sample Ratio Mismatch, Causal Inference, Uplift Modeling, T-Learner, X-Learner, Treatment-Effect Analysis

MLOps & Deployment: MLflow, ZenML, FastAPI, Docker, GitHub Actions, pytest, Model Registry, Model Monitoring, Drift Monitoring, Automated Retraining

Data & Applications: MySQL, MongoDB, Redis, Streamlit, REST APIs, Git, GitHub, Plotly, Matplotlib

PROJECTS

LocalMart AI — Geo-Aware Retail Promotion Intelligence

Python | CatBoost | Causal ML | PuLP | Streamlit | Statistics | Optimization

- Built an end-to-end retail decision system combining demand forecasting, promotion uplift modeling, local GTA market context, product economics, inventory, and constrained optimization across **15 stores, 300 products, and 104 weeks** of semi-synthetic retail history.
- Developed a CatBoost demand model achieving **3.09-unit test MAE** and a **29.71% validation MAE improvement** over a lag-1 baseline.
- Built a T-Learner uplift model achieving **0.96-unit uplift MAE** and **0.828 correlation** with known simulated treatment effects, with the top 10% of ranked products achieving **2.17× uplift enrichment**.
- Formulated a PuLP integer-optimization layer respecting flyer-space, category, budget, inventory, and profitability constraints; simulated evaluation produced **\$20.6K incremental profit** with **97.1% positive-profit selections**.
- Developed an interactive Streamlit application communicating store-level recommendations, forecasting performance, treatment effects, unit economics, and optimization results.

Movie Recommendation A/B Testing & Uplift Modeling

Python | Experimentation | Statistics | Causal Inference | Uplift Modeling

- Designed and analyzed a randomized product experiment across **28,000 synthetic users** comparing personalized recommendations with a popularity-based control strategy.
- Detected a randomization defect caused by reused RNG state, corrected treatment assignment, and validated the redesigned experiment using **sample-ratio mismatch and covariate-balance checks**.
- Measured CTR of **9.343% for control vs. 10.188% for treatment**, corresponding to **+0.846 percentage-point lift**, **p = 0.0171**, and a **95% CI of +0.150 to +1.541 pp**.
- Identified heterogeneous treatment effects: returning users showed **+1.595 pp lift**, while new users showed **-0.472 pp lift**; interaction analysis measured an additional **+2.066 pp treatment effect for returning users (p = 0.0037)**.
- Compared T-Learner and X-Learner uplift models using CATE error, Uplift@K, and Qini analysis and recommended targeted personalization for returning users rather than a full rollout.

End-to-End House Price Prediction & MLOps System

Python | scikit-learn | ZenML | MLflow | FastAPI | Docker | CI/CD

- Developed a production-oriented ML lifecycle for Ames Housing covering training, experiment tracking, model registration, serving, monitoring, retraining, model promotion, and rollback.
- Built a regression pipeline achieving **\$13,044.81 MAE**, **\$18,626.84 RMSE**, and **0.9221 R²** on the test dataset.
- Implemented ZenML training pipelines, MLflow experiment tracking and model registry, FastAPI inference endpoints, Pydantic validation, Docker containerization, and Prometheus-compatible operational metrics.
- Added drift and performance monitoring with policy-driven retraining and champion-versus-candidate promotion gates, including rollback to a previous champion model.
- Implemented GitHub Actions CI with linting, Docker validation, and **28 automated tests**.

Flight Pulse — Flight Analytics & Grounded AI Analyst

Python | SQL | MySQL | Streamlit | APIs | Gemini | Data Engineering

- Built an end-to-end flight analytics application integrating operational flight data, hourly weather context, disruption-news metadata, SQL analytics, a Streamlit dashboard, and a grounded AI analyst.
- Retrieved and normalized **72 real Toronto Pearson (YYZ) flights** and **48 hourly weather observations**, matching weather context to **72/72 flight records** by scheduled UTC hour.
- Designed parameterized MySQL persistence and reusable analytics for flight status, airline, route, delay, time, and data-quality analysis.
- Implemented **7 approved read-only analyst tools** for Gemini function calling, preventing unrestricted model-generated SQL and validating tool names and arguments before query execution.
- Built **70 automated tests** covering analytical logic and external-service boundaries while using mocked integrations to avoid unnecessary API consumption.

EDUCATION

Conestoga College — Waterloo, Ontario

Diploma, Computer Programming | 2024–2026

Relevant areas: Programming, Data Structures & Algorithms, Object-Oriented Programming, Databases, Web Development, Software Quality Assurance

ADDITIONAL EXPERIENCE

Defron Security Services — Client Services / Security

Mississauga / Greater Toronto Area, Ontario | [CONFIRM EMPLOYMENT DATES]

- Supported clients and visitors, responded to inquiries and concerns, and maintained professional communication in fast-paced customer-facing environments.
- Developed communication, problem-solving, situational awareness, and responsibility while working independently and with operational teams.

Winners — Sales Representative

Toronto, Ontario | [CONFIRM EMPLOYMENT DATES]

- Assisted customers with product inquiries and transactions while supporting merchandise organization, stocking, and store operations.
- Worked collaboratively in a high-volume retail environment requiring attention to detail, time management, and customer service.

Conestoga College — Volunteer Mentor

Waterloo, Ontario | [CONFIRM VOLUNTEER DATES]

- Provided individual and small-group support to students and helped communicate academic and organizational information clearly.